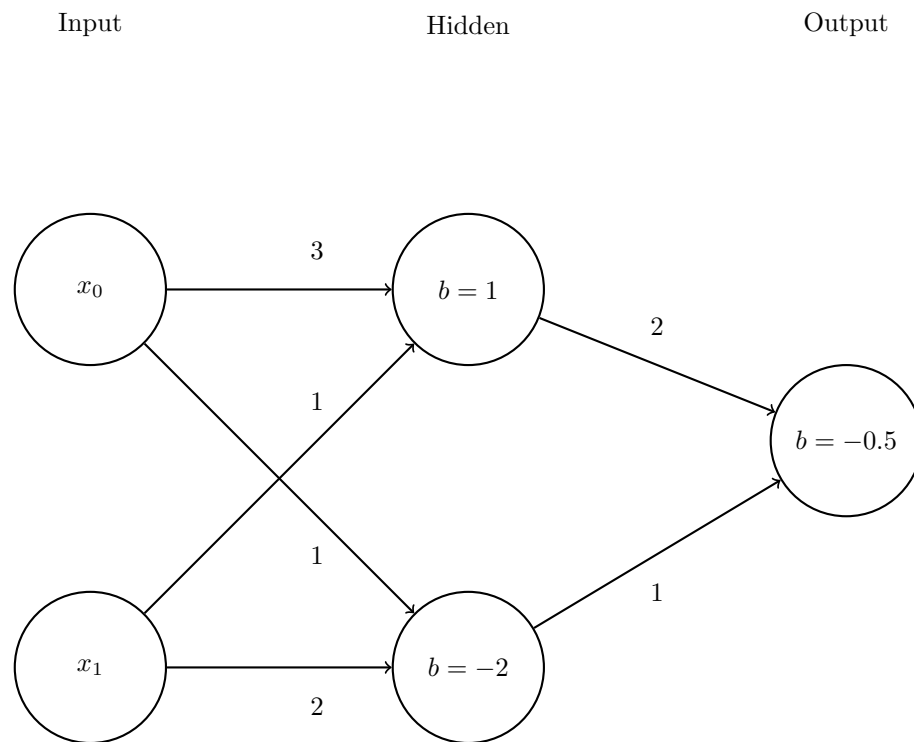


1 Activation Function

$$activation(x) = \begin{cases} 1, & \text{if } x \geq 0.5 \\ 0, & \text{if } x < 0.5 \end{cases}$$

2 Network



3 Forward

3.1 Hidden Layer Calculation

- $hidden = activation((W_{\text{hidden}}^T * inputs) + bias)$
- $hidden = activation((\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} * \begin{bmatrix} x_0 \\ x_1 \end{bmatrix}) + \begin{bmatrix} 1 \\ -2 \end{bmatrix})$ Here: $W_{\text{hidden}} = W_{\text{hidden}}^T$
- $hidden = activation(\begin{bmatrix} 3 * x_0 + 1 * x_1 \\ 1 * x_0 + 2 * x_1 \end{bmatrix} + \begin{bmatrix} 1 \\ -2 \end{bmatrix})$
- **Insert Values:** $x_0 = 1, x_1 = 1$
- $hidden = activation(\begin{bmatrix} 3 * 1 + 1 * 1 + 1 \\ 1 * 1 + 2 * 1 - 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix})$
- $hidden = activation(\begin{bmatrix} 5 \\ 1 \end{bmatrix})$
- $hidden = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

3.2 Output Layer Calculation

- $output = activation((W_{\text{output}}^T * hidden) + bias)$
- $W_{\text{output}} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$
- $W_{\text{output}}^T = [2 \quad 1]$
- $output = activation((\begin{bmatrix} 2 & 1 \end{bmatrix} * \begin{bmatrix} 1 \\ 1 \end{bmatrix}) - 0.5)$
- $output = activation((\begin{bmatrix} 2 * 1 + 1 * 1 \end{bmatrix}) - 0.5)$
- $output = activation(2.5)$
- $output = 1$

3.3 All inputs

x_0	x_1	Expected	Result
1	1	1	1
2	1	1	1
3	2	0	1
3	-2	1	1

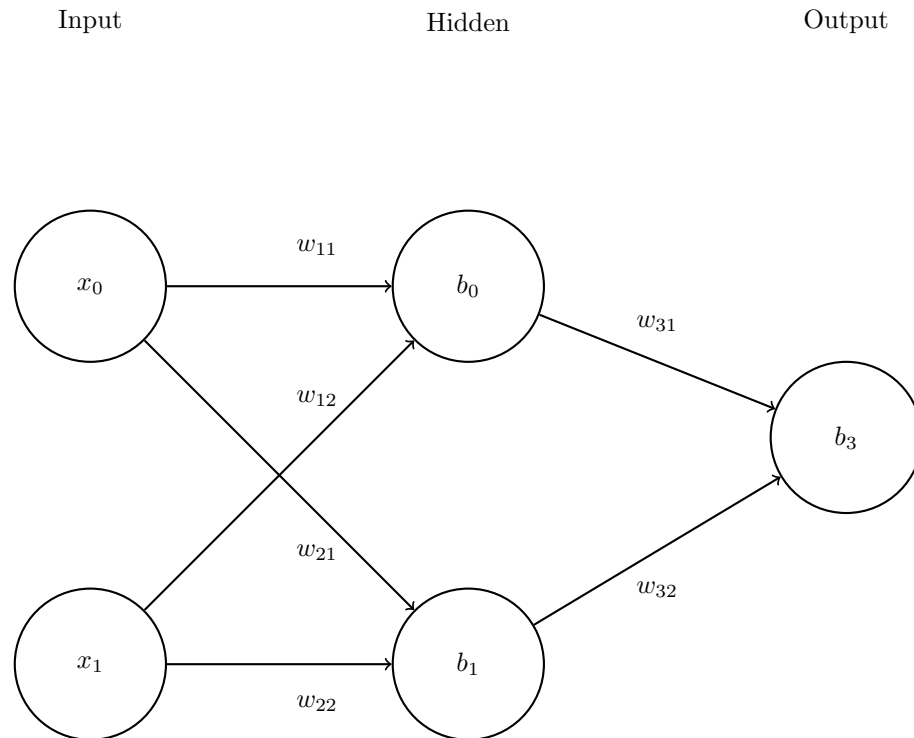
For $x_0 = 3$ and $x_1 = 2$, the classification is wrong!

4 Notes

- The number of columns in the weight matrix of a layer always equals the number of neurons in the layer.
- The number of rows equals the number of neurons in the previous layer.

In our case, this means that the hidden layer is represented by a 2×2 matrix, and the output layer by a 2×1 matrix.

4.1 General



4.2 Hidden Layer Calculation

- $hidden = activation((W_{\text{hidden}}^T * inputs) + bias)$
- $hidden = activation((\begin{bmatrix} w11 & w21 \\ w12 & w22 \end{bmatrix}^T * \begin{bmatrix} x_0 \\ x_1 \end{bmatrix}) + \begin{bmatrix} b_0 \\ b_1 \end{bmatrix})$
- $\begin{bmatrix} w11 & w21 \\ w12 & w22 \end{bmatrix}^T = \begin{bmatrix} w11 & w12 \\ w21 & w22 \end{bmatrix}$
- $hidden = activation(\begin{bmatrix} w11 * x_0 + w12 * x_1 \\ w21 * x_0 + w22 * x_1 \end{bmatrix} + \begin{bmatrix} b_0 \\ b_1 \end{bmatrix})$

4.3 Output Layer Calculation

- $output = activation((W_{\text{output}}^T * hidden) + bias)$
- $W_{\text{output}} = \begin{bmatrix} w31 \\ w32 \end{bmatrix}$
- $W_{\text{output}}^T = \begin{bmatrix} w31 & w32 \end{bmatrix}$
- $output = activation((\begin{bmatrix} w31 & w32 \end{bmatrix} * \begin{bmatrix} w31 \\ w32 \end{bmatrix}) - b_3)$